Preferences and indifference curves

Lecture notes

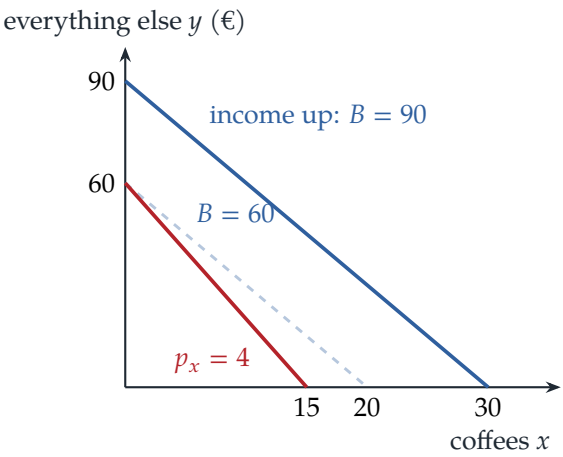
What this lecture is about. Last lecture described what a consumer *can* buy: the budget set. Today we describe what a consumer *wants*. We do this without any formula. We start from four simple rules about how people rank bundles, and from those rules we draw the indifference curves. They show how a consumer trades one good against another. Then we ask what we can learn about preferences just by watching choices on budget lines.

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1 The budget set, briefly

A simple example. Each month you spend a budget of $B = €60$ on coffees (x, at $p_x = €3$ each) and on everything else (y, the composite good, measured in euro, so $p_y = 1$). The budget line is

$$3x + y = 60 \qquad \Longleftrightarrow \qquad y = 60 - 3x.$$



- **The two ends.** All of the budget on coffees gives $B/p_x = 60/3 = 20$ coffees. All of it on everything else gives $B/p_y = €60$.
- **Income rises** to $B = €90$: both ends move out (30 coffees, €90), the slope stays -3. The line *shifts* out, parallel to itself.
- **The coffee price rises** to $p_x = €4$: the x-end moves in to $60/4 = 15$, the y-end stays at €60. The line *rotates* around the y-end and becomes steeper.
- **The composite good.** Because y is money for everything else, $p_y = 1$. The y-end is then the budget itself, and the slope is simply $-p_x$: each coffee costs €3 of everything else.

What is the slope of the budget line called?

The size of the slope, p_x/p_y, is the **relative price** of x: how many units of y the *market* makes you give up for one more unit of x. It is the opportunity cost of x in units of y. Some books call it the *market rate of exchange*.

It is **not** the same as how much you would be *willing* to give up; that is a different idea (section 4, on trade-offs).

2 Preferences and four rules

2.1 Bundles and how we compare them

A **bundle** $A = (x_A, y_A)$ lists an amount of each good, for example 8 coffees and €36 of everything else. A consumer's **preferences** are her ranking of all bundles. We only ask which of two bundles she likes more; we never ask *how much* more.

Notation

$A \succsim B$	A is **at least as good as** B	the basic relation ("weakly preferred")
$A \succ B$	A is **strictly preferred** to B	$A \succsim B$, but not $B \succsim A$
$A \sim B$	the consumer is **indifferent**	$A \succsim B$ and $B \succsim A$

Everything starts from the single symbol $\succsim$. Strict preference and indifference are both defined from it.

Economists assume that preferences follow four rules. The first two are about consistency. The last two are about the goods themselves.

2.2 Rule 1: completeness

Completeness

For any two bundles A and B, at least one of $A \succsim B$ or $B \succsim A$ holds. The consumer can always compare.

"I can't decide" is not allowed. Note the difference with indifference: an indifferent consumer *has* compared the two bundles and found them equally good. Completeness rules out the consumer who refuses to compare at all.

2.3 Rule 2: transitivity

Transitivity

If $A \succsim B$ and $B \succsim C$, then $A \succsim C$.

Transitivity says the ranking hangs together: it has no cycles. It looks obvious, but plenty of everyday relations are not transitive. Team 1 can beat team 2 and team 2 can beat team 3, while team 3 still beats team 1. Your friend's friend need not be your friend.

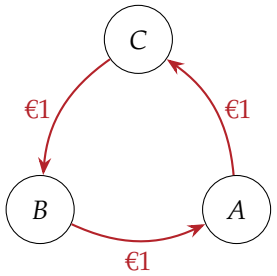
Example: the money pump

Suppose you rank three bundles A, B and C in a cycle:

$$A \succ B, \qquad B \succ C, \qquad C \succ A.$$

You hold bundle C. A seller offers to switch you to B for a fee of €1. You prefer B to C, so you pay. Then the seller offers A for another €1: you prefer A to B, so you pay again. Then the seller offers C for €1: you prefer C to A, so you pay a third time.

After three swaps you hold exactly the bundle you started with, and you are €3 poorer. The seller can repeat this forever. A person with cyclic preferences can be "pumped" until the money is gone. This is the classic argument for transitivity: people with intransitive preferences would lose money to anyone who noticed.



each arrow: a swap you are happy to pay for

Insight: where intransitivity comes from in real life

Real people do sometimes choose in cycles. A common reason:

- **Many considerations, counted like votes.** If you judge options on price, quality and speed, and pick the option that wins on two of the three, you can end up in a cycle even though each single consideration ranks the options consistently. The same happens to groups: three voters, each with a perfectly transitive ranking, can produce a cycle under pairwise majority voting.

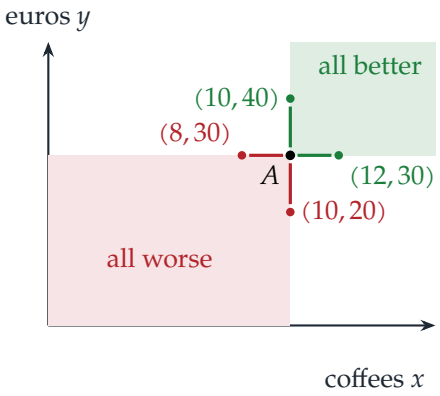
The theory keeps transitivity as an assumption. Section 7 shows how we can *test* it with data.

2.4 Rule 3: more is better

More is better (monotonicity)

If bundle A has at least as much of each good as B, and more of at least one, then $A \succ B$.

Start at $A = (10, 30)$: 10 coffees and €30. Two more coffees and the same money, $(12, 30)$, is better. The same coffees and €10 more, $(10, 40)$, is better too. So is everything up and to the right of A: more of one good and no less of the other. In the same way $(8, 30)$ and $(10, 20)$ are worse, and so is everything down and to the left.



This rule is about *goods*: things you want. It fails for *bads*, such as noise, pollution or spam messages, where more is worse. With a bad, the rule must be turned around (less of the bad is better). It can also fail when you are full (satiated): ten slices of pizza may be worse than nine. In this course both goods are goods unless we say otherwise.

2.5 Rule 4: convexity

Convexity

Take two equally good bundles, $A \sim B$. Then any mix of them is at least as good as either one:

$$tA + (1 - t)B \succsim A \qquad \text{for every } t \text{ between 0 and 1.}$$

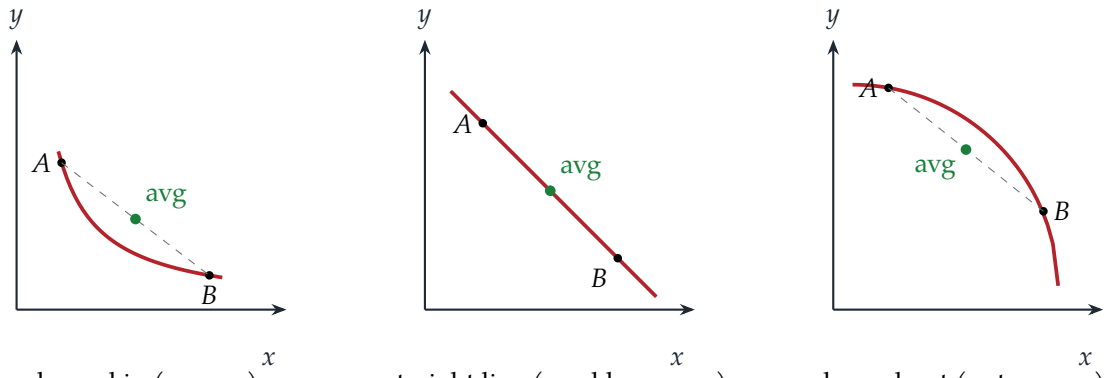
A **mix** takes a share t of bundle A and a share $1 - t$ of bundle B. The half-half mix ($t = \frac{1}{2}$) is simply the **average** of the two bundles. If the mix is always *strictly* better (for every t strictly between 0 and 1, with $A \neq B$), preferences are **strictly convex**.

Averages beat extremes. Suppose you are indifferent between a week of eating only pizza and a week of eating only salad. Convexity says that a week with half pizza and half salad is at least as good as either extreme. Most people like variety: the first salad after six pizza days is worth a lot.

The same idea in numbers. Take two bundles you like equally: $A = (2, 50)$, 2 coffees and €50 of everything else, and $B = (14, 14)$. Their average is

$$\tfrac{1}{2}A + \tfrac{1}{2}B = \left(\tfrac{2+14}{2}, \tfrac{50+14}{2}\right) = (8, 32).$$

Convexity says that $(8, 32)$ is at least as good as A and B. In a picture, the average lies at the midpoint of the straight segment between A and B.



bowed in (convex): the average is *better*

straight line (weakly convex): the average is *equally good*

bowed out (not convex): the average is *worse*

Insight: does the curve have to be bowed in? No

Convexity is an *assumption*, not a law of nature. Three things are worth knowing.

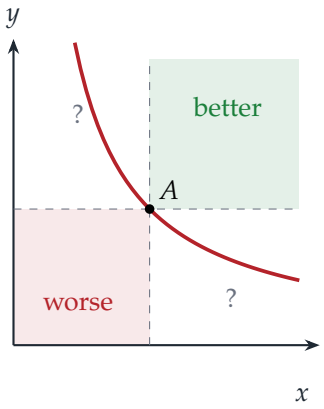
1. **A straight line is allowed.** The definition says the mix is *at least as good*. On a straight curve of equally good bundles the mix lies on the same curve, so it is exactly as good. That satisfies (weak) convexity. Only *strict* convexity fails. Perfect substitutes (section 5, on substitutes and complements) have straight-line curves and are weakly convex.
2. **What convexity really rules out** is a curve bowed *away* from the origin. There the extremes beat the average: you would always want to spend everything on one good. That does happen: you may be indifferent between painting your room all red and all blue, but a half-half mix gives a purple you dislike: worse than either.
3. **Why we like the assumption.** Convexity is what makes the analysis simple. Section 6, on corners, shows why, once the budget line is back next to these curves.

3 Indifference curves

3.1 Building the curve from the rules

Pick any bundle A. The rules already tell us a lot about which bundles are better or worse than A.

- Every bundle up and to the right of A has more of at least one good and no less of the other: by *more is better*, it is **better** than A.
- Every bundle down and to the left of A is **worse** than A, for the same reason.
- So the bundles that are exactly as good as A can only lie in the other two areas: up and to the left (more y, less x), or down and to the right (more x, less y).



Join all bundles that are exactly as good as A. The result is the **indifference curve** through A, and it must slope downward.

Indifference curve

An **indifference curve** is the set of all bundles that the consumer likes exactly as much as a given bundle. A family of such curves, one through every bundle, is the **indifference map**. Curves further from the origin are better.

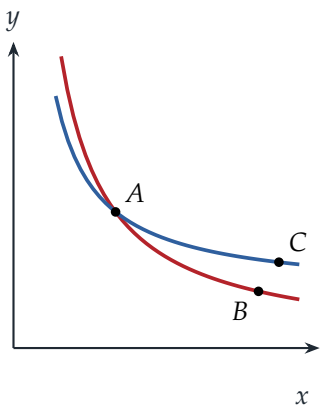
3.2 Five properties, and which rule gives each

property of indifference curves	comes from
1. there is a curve through every bundle	completeness
2. curves slope downward	more is better
3. curves further out are better	more is better
4. curves never cross	transitivity *and* more is better
5. curves are bowed toward the origin	convexity

An upward-sloping curve? If an indifference curve sloped upward, a bundle further up the curve would have more of *both* goods and yet be only equally good. That breaks *more is better*, provided both are goods. (If y were a bad, such as minutes of commuting, an upward slope makes sense: you need more coffees to put up with a longer commute.)

3.3 Why indifference curves cannot cross

Suppose two curves cross at bundle A, as in the figure. Bundle B is on the first curve, bundle C on the second, and C lies up and to the right of B.

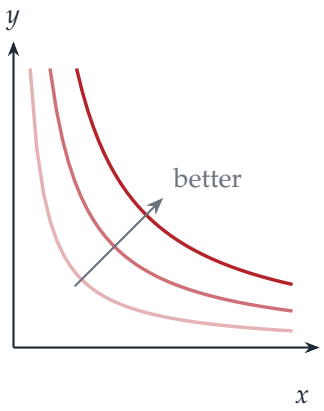


- $A \sim B$ (same curve), and $A \sim C$ (same curve).
- By *transitivity*, $B \sim C$.
- But C has more of both goods than B (it lies up and to the right), so by *more is better*, $C \succ B$.

Both cannot be true. So two indifference curves can never cross. Notice that we needed *two* rules for this argument.

3.4 The indifference map

Every bundle has its own curve, so the plane is filled with curves, like the contour lines of a hiking map. Picture utility as a mountain over the floor of bundles: a flat plane that cuts the mountain at one height traces one indifference curve, and seen from above the cuts at several heights are the contour lines $u_1 < u_2 < u_3$. Each curve is a "height" of satisfaction. Moving to a curve further out means moving to a better bundle. The map is a complete picture of the consumer's preferences. Two people with different tastes have different maps: someone who loves coffee has steep curves (she gives up a lot of other spending for one more coffee), someone who cares little about them has flat curves. In numbers: on the coffee lover's curve, going from 9 coffees and €13 to 10 coffees and €10 means giving up €3 for the 10th coffee. On the other person's curve, going from 9 coffees and €10.50 to 10 coffees and €10, the 10th coffee costs only €0.50. Both end at the same bundle $(10, €10)$, yet the 10th coffee is worth €3 to one and €0.50 to the other: that difference is the steepness.



4 Trade-offs: the marginal rate of substitution

4.1 How much would you give up?

Indifference curves show trade-offs. Start at a bundle and ask: *how much of everything else* (y) *would you give up for one more coffee* (x)*, and still be equally happy?* The answer is the marginal rate of substitution.

Marginal rate of substitution (MRS)

The **MRS** at a bundle is the amount of y the consumer is willing to give up for one more unit of x while staying on the same indifference curve. In the picture it is the steepness of the indifference curve at that bundle: the slope without its minus sign.

The MRS is a *willingness*: it is inside the consumer's head. The relative price from section 1, on the budget set, is what the *market* asks. Keep the two apart.

Insight: your rate and the market's rate are both Δy over Δx

Every slope is rise over run: $\Delta y/\Delta x$, with y on top and x at the bottom. Both rates answer the same question, *how much y goes for one more x?*

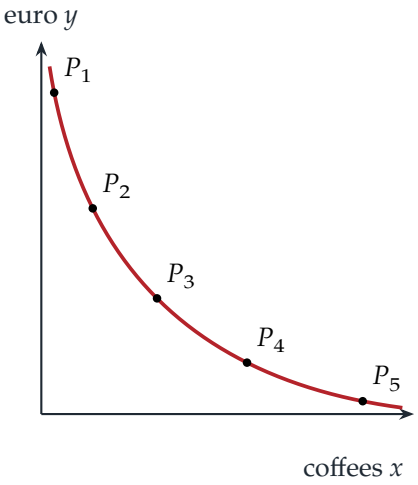
- **The market (budget line).** One more coffee $(\Delta x = +1)$ costs p_x euros, so you have p_x/p_y units less of y: $\Delta y = -p_x/p_y$. The slope is $\Delta y/\Delta x = -p_x/p_y$; with $p_y = 1$ it is just $-p_x$.
- **You (indifference curve).** One more coffee is worth some amount of y to you: MRS $= -\Delta y/\Delta x$ along your curve.

Check with numbers: giving up €9 for 3 more coffees means $-\Delta y/\Delta x = 9/3 = 3$ euros per coffee. The upside-down ratio $\Delta x/\Delta y = 1/3$ would be coffees per euro: the same trade-off, but not the MRS.

4.2 Diminishing MRS

Here are five bundles on one indifference curve (coffees, euro of everything else):

bundle	coffees x	euro y	€ per extra coffee
P_1	1	25	
P_2	4	16	$9/3 = 3$
P_3	9	9	$7/5 = 1.4$
P_4	16	4	$5/7 \approx 0.71$
P_5	25	1	$3/9 \approx 0.33$

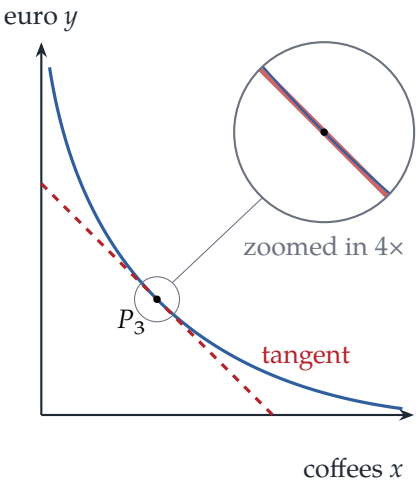


With few coffees and lots of money ($P_1 \to P_2$), you give up €3 for each extra coffee. With many coffees and little money ($P_4 \to P_5$), you give up only about €0.33 for each. The more of a good you already have, the less of the other good you will give up for one more unit of it: your willingness to pay for one more coffee, in euros of everything else, falls. These numbers are averages over each step; the MRS at the points themselves (the tangent slopes, section 4.3) is 5 at P_1, 2 at P_2, 1 at P_3, 0.5 at P_4 and 0.2 at P_5. This is a **diminishing MRS**, and it is exactly what makes the curve bowed toward the origin: steep on the left, flat on the right. Convexity and a diminishing MRS are two descriptions of the same shape.

4.3 The MRS at a point: zoom in until the curve is straight

The numbers in the table are averages over a stretch of the curve. The MRS *at* one bundle is the steepness of the curve right there. How do we measure the steepness of a curve? We zoom in.

Look at $P_3 = (9, 9)$ through a magnifying glass. The closer you look, the straighter the curve looks. Very close up, it is almost the same as the straight line that just touches it at P_3: the **tangent line**. The MRS at P_3 is the steepness of that tangent. On this curve the tangent at P_3 has slope -1, so the MRS at P_3 is 1: around 9 coffees and €9, one more coffee is worth about €1 to you.
This is how mathematics deals with curves: replace the curve by its tangent line near the point you care about (**linearisation**). The slope of the tangent is what a derivative measures.



Insight: how we get the slope at one point: shrink the step

Start at $P_3 = (9, 9)$ and take steps of different sizes along the curve. For each step, the trade-off per coffee is $-\Delta y/\Delta x$:

step	Δx	Δy	per coffee, $-\Delta y/\Delta x$
to $(16, 4)$	7	-5	0.71
to $(12, 6.43)$	3	-2.57	0.86
to $(10, 8.05)$	1	-0.95	0.95
to $(9.1, 8.90)$	0.1	-0.099	0.99

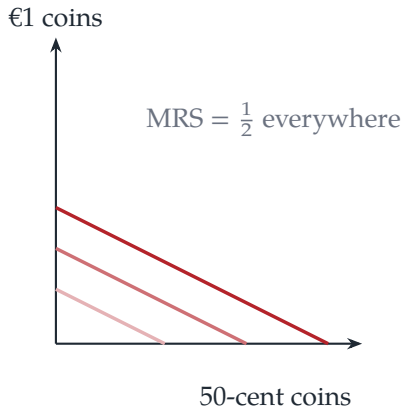
As the step shrinks, the number settles at 1. That limit is the slope of the tangent at P_3 (without the minus sign): the MRS at P_3 is 1. Next lecture we get the same number directly from a utility function, with partial derivatives, holding utility constant.

5 Perfect substitutes and perfect complements

Two extreme kinds of preferences are useful benchmarks. We can describe both with the trade-off question alone, without any formula: *how much y would you give up for one more x?*

5.1 Perfect substitutes: the answer never changes

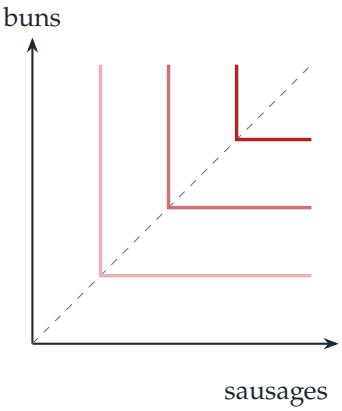
Two goods are **perfect substitutes** if they do the same job at a fixed rate. Think of 50-cent coins (x) and €1 coins (y). Two 50-cent coins do exactly what one €1 coin does, and you only care about how much money you have. Then one more 50-cent coin is always worth half a €1 coin to you, whether you have many 50-cent coins or none.
The MRS is **constant**, so the indifference curves are **straight parallel lines**. The rate does not have to be one for one: here it is 2 to 1.



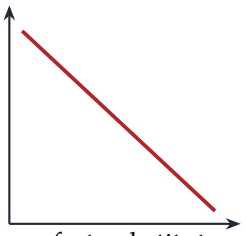
5.2 Perfect complements: no swapping at all

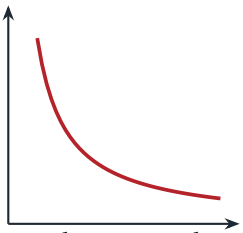
Two goods are **perfect complements** if they are used in fixed proportions. A hot dog needs one sausage and one bun. Five sausages and five buns give you five hot dogs. A sixth sausage without a sixth bun gives you nothing extra; so does a sixth bun without a sixth sausage.

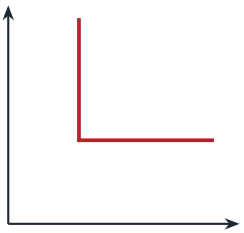
The indifference curves are **L-shaped**, with the corner on the line "as many buns as sausages". On the flat and vertical arms, more of one good alone does not help. Here "more is better" only holds in the weak form: more of *both* is better. On the flat arm the MRS is 0: an extra sausage alone is worth no buns. On the vertical arm the curve is vertical. At the corner the slope jumps, so there is no single MRS there.



5.3 One scale from "swap freely" to "never swap"


perfect substitutes:
straight lines,
swap rate fixed


ordinary goods:
bowed-in curves,
swap rate falls


perfect complements:
L-shapes, no swapping

Both extremes are only *weakly* convex: a mix of two equally good bundles is always at least as good, but for some pairs it is only *equally* good. For substitutes that is any two points on the same line; for complements, any two points on the same arm of the L. So straight lines and L-shapes do not violate convexity; they fail only *strict* convexity.

Insight: how to tell them apart without formulas

Ask the trade-off question at several bundles. If the answer is "always the same amount", the goods are perfect substitutes and the curves are straight lines. If the answer is "nothing, unless I also get more of the other good", they are perfect complements and the curves are L-shapes. Everything else is in between: the answer depends on how much you already have, and the curves bow in.

6 Corners, and a first look at utility

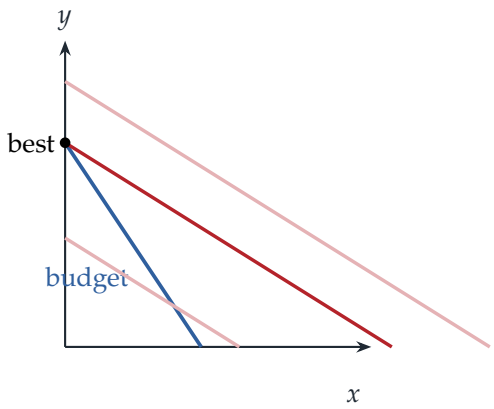
Remember perfect substitutes (section 5, the 50-cent and €1 coins): their indifference curves are straight lines. That is the classic case of a corner.

Next lecture we bring the budget line back and look for the best affordable bundle. One idea already works without any formula: compare the two rates.

- The **MRS**: how many euro of everything else you are *willing* to give up for one more coffee.
- The **relative price** p_x/p_y: how many euro the market *makes* you give up (€3 per coffee in our model).

If at your current bundle the MRS is larger than the price, one more coffee is worth more to you than it costs: buy more coffees. If the MRS is smaller, buy fewer. You stop when the two are equal, or when you cannot move any further because you have reached an end of the budget line.

Perfect substitutes almost always end in a corner. The MRS never changes, so it is either above the relative price everywhere (spend everything on x) or below it everywhere (spend everything on y). The best bundle is an end of the budget line: a **corner solution**. In the figure the indifference lines are flatter than the budget line, so the consumer ends at the y-end.



Corners are not only for perfect substitutes. Ordinary bowed-in curves give a corner too if their MRS stays on one side of the price along the whole budget line.

Why convexity makes life simple. With bowed-in curves, the best affordable bundle is usually in the interior of the budget line and unique, and it moves smoothly when prices change; next lecture this gives smooth demand curves. Someone who dislikes mixing (the red-or-blue paint case of section 2) has curves bowed *away* from the origin and always ends at an end of the budget line.

Insight: the order of ideas

First your own rate (the MRS), then the market's rate (the relative price), then compare them. Corners then need no new idea: they are what happens when the comparison never balances.

6.1 A first look at utility

Utility

A **utility function** gives each bundle a number, so that a better bundle gets a higher number: $u(A) > u(B)$ exactly when $A \succ B$, and $u(A) = u(B)$ exactly when $A \sim B$.

Take three bundles at the same number of coffees, one on each of three indifference curves: $A = (10, 6)$, $B = (10, 12)$ and $C = (10, 24)$. More euros, so $C \succ B \succ A$. Now label them with three different utility functions:

bundle	u	$2u + 5$	u^2
A	1	7	1
B	2	9	4
C	3	11	9
order	$A < B < C$	$A < B < C$	$A < B < C$
$C \div A$	3	≈ 1.6	9

- **Ordinal:** every column gives the same order, so every column describes the same preferences and the same indifference curves. (Squaring keeps the order here only because all the u values are positive.) Only the order carries information.
- **Cardinal** would mean that the numbers themselves say something, such as "C is three times as good as A". But that answer is 3, about 1.6 or 9, depending only on which labels we picked. So we do not read utility numbers that way.

Next lecture: utility functions as formulas, marginal utility, and the best affordable bundle.

7 What choices reveal

7.1 From choices to preferences

Choices are not preferences. A *preference* is what you like; it is inside you and nobody can see it. A *choice* is what you actually pick, and it depends on two things: what you like *and* what you can afford. In everyday language, and in many languages, the two words are used almost interchangeably ("my choice", "my preference"). In economics we keep them apart, because the budget sits between them: the same preferences lead to different choices when prices or income change.

So far we assumed that we know a consumer's preferences. In practice we only see what people *choose*. Can we check the four rules from choices alone? Yes, and this is a large part of modern experimental economics.

In a budget experiment (Choi, Fisman, Gale and Kariv, 2007, and many studies since), each participant sees a series of straight budget lines, one at a time, and picks one point on each line. The lines are drawn at random and cross each other often. The researcher sees nothing but the lines and the chosen points.

It helps to write every line in the intercept form of lecture 5,

$$\frac{x}{X} + \frac{y}{Y} = 1,$$

where X and Y are the two ends. The left-hand side is then the **cost of a bundle as a share of the budget**: a bundle with $\frac{x}{X} + \frac{y}{Y} < 1$ is affordable on that line with money to spare.

Directly revealed preferred

If a consumer chose bundle A on a line where bundle B was also affordable, then A is **directly revealed preferred** to B. She could have had B and took A. If B was strictly cheaper than the budget, A is revealed *strictly* preferred: with more is better, she could have had B plus some extra money.

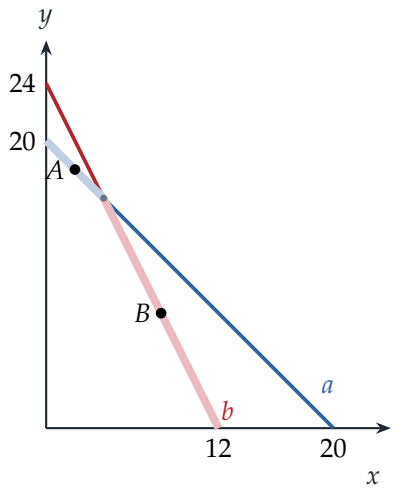
7.2 Two lines: consistent or not

Take line a with ends $(20, 20)$, that is $\frac{x}{20} + \frac{y}{20} = 1$, and line b with ends $(12, 24)$. The two lines cross at $(4, 16)$.

A contradiction. On line a she picks $A = (2, 18)$. On line b she picks $B = (8, 8)$.

- Was B affordable on line a? Its cost share is $\frac{8}{20} + \frac{8}{20} = 0.8 < 1$. Yes, so A is revealed preferred to B.
- Was A affordable on line b? $\frac{2}{12} + \frac{18}{24} \approx 0.92 < 1$. Yes, so B is revealed preferred to A.

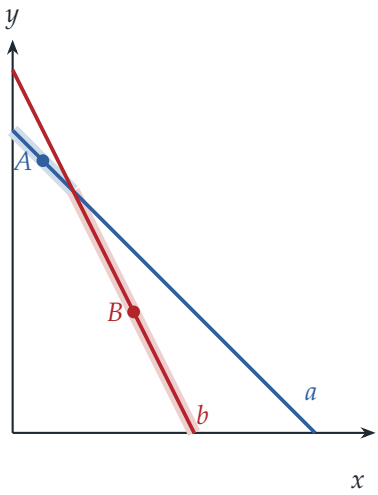
Each choice was affordable when the other was made. No consumer with preferences that follow our rules chooses like this. This is a violation of the **weak axiom of revealed preference** (WARP). It can only happen if both choices sit on the "inner" pieces of the two lines, the pieces that lie below the other line.



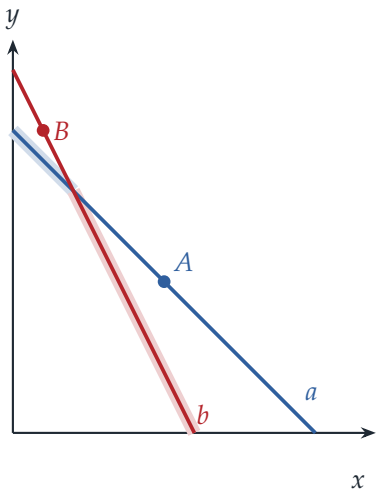
shaded: the inner pieces

7.3 All four cases on the same two lines

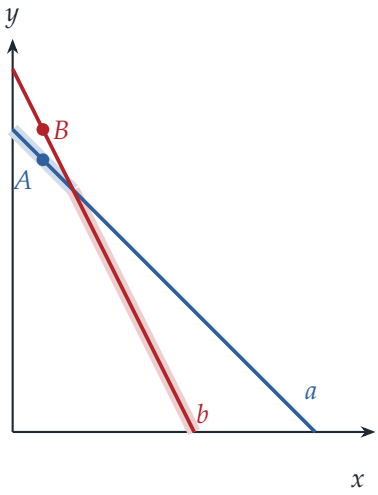
Whether two choices contradict each other depends only on *where* on the two lines they sit. Each line has an **inner piece** (the part below the other line, shaded) and an **outer piece** (the part above the other line). A bundle on an inner piece is affordable on the other line; a bundle on an outer piece is not. That gives four cases.



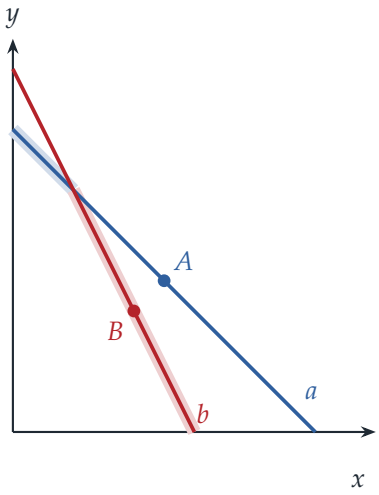
1 · both inner. Each was affordable when the other was picked: a contradiction.



2 · both outer. Neither was affordable on the other line. Nothing is revealed: consistent.



3 · A inner, B outer. A was affordable on line b, so B is revealed preferred to A, and nothing the other way: consistent.



4 · A outer, B inner. B was affordable on line a, so A is revealed preferred to B, and nothing the other way: consistent.

shaded: the inner pieces (the part of each line that lies below the other line)

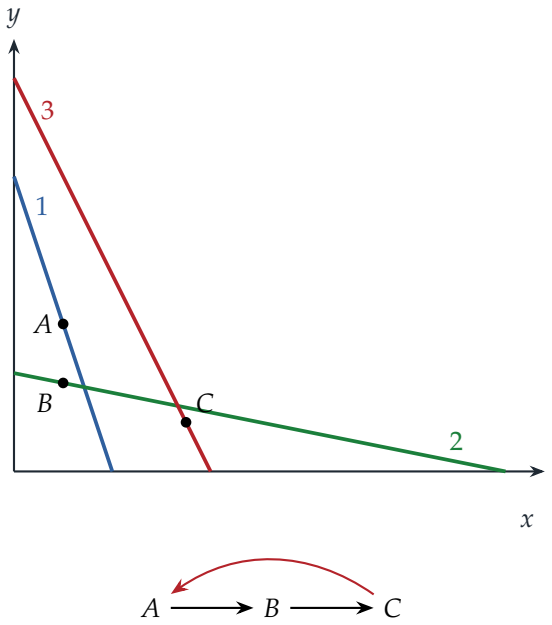
case	A on line a	B on line b	B affordable on a?	A affordable on b?	verdict
1	$(2, 18)$	$(8, 8)$	$0.8 < 1$: yes	$0.92 < 1$: yes	contradiction
2	$(10, 10)$	$(2, 20)$	$1.1 > 1$: no	$1.25 > 1$: no	consistent
3	$(2, 18)$	$(2, 20)$	$1.1 > 1$: no	$0.92 < 1$: yes	consistent
4	$(10, 10)$	$(8, 8)$	$0.8 < 1$: yes	$1.25 > 1$: no	consistent

The numbers are the **cost shares**: $\frac{x}{20} + \frac{y}{20}$ on line a and $\frac{x}{12} + \frac{y}{24}$ on line b. Only case 1 breaks the rules, and only because *both* picks sit on the inner pieces. So the rule you can see on a graph is: **two choices contradict when each lies below the other line** (or when one sits exactly where the lines cross and the other below).

7.4 Three lines: a cycle

With more lines, revealed preferences can be chained: if A is revealed preferred to B and B to C, then A is *indirectly* revealed preferred to C. This is transitivity again. A **cycle** (A over B over C over A) means the choices cannot come from any transitive ranking. The **generalised axiom of revealed preference** (GARP) says: no such cycles with a strict step.

line	ends (X, Y)	choice
1	$(20, 60)$	$A = (10, 30)$
2	$(100, 20)$	$B = (10, 18)$
3	$(40, 80)$	$C = (35, 10)$



- On line 1, B costs $\frac{10}{20} + \frac{18}{60} = 0.8$ of the budget: A revealed preferred to B.
- On line 2, C costs $\frac{35}{100} + \frac{10}{20} = 0.85$: B revealed preferred to C.
- On line 3, A costs $\frac{10}{40} + \frac{30}{80} = 0.625$: C revealed preferred to A.
- Also on line 3, B costs $\frac{10}{40} + \frac{18}{80} \approx 0.48$: C revealed preferred to B as well.

Insight: why A and B alone are fine, but the chain is not

Take only lines 1 and 2. B was affordable on line 1 (cost share 0.8), so A is revealed preferred to B. But A was *not* affordable on line 2: its cost share there is $\frac{10}{100} + \frac{30}{20} = 1.6 > 1$. So there is no "$B$ over A" and no contradiction: this is case 4 of section 7.3 (A on the outer piece, B on the inner piece).

The point $K \approx (14.3, 17.1)$ where lines 1 and 2 cross is on both lines, so it was affordable both times: A is revealed at least as good as K, and B is revealed at least as good as K. That is no problem either: two bundles that both beat K say nothing about each other. (This is only true while we look at lines 1 and 2. Once line 3 comes in, A and B clash too, through C.)

A contradiction needs one bundle over another *and* the other way round. Three lines can deliver that through a chain: A over B (line 1), B over C (line 2), C over A (line 3), so A is over C and C is over A at the same time. (Line 3 also puts C over B, which clashes directly with line 2.)

Here is every check at once: on each line, the cost share of the two bundles that were *not* chosen there.

line (choice)	bundle	cost share	affordable?
1 (picked A)	$B = (10, 18)$	$\frac{10}{20} + \frac{18}{60} = 0.8$	yes: A over B
	$C = (35, 10)$	$\frac{35}{20} + \frac{10}{60} \approx 1.92$	no
2 (picked B)	$A = (10, 30)$	$\frac{10}{100} + \frac{30}{20} = 1.6$	no
	$C = (35, 10)$	$\frac{35}{100} + \frac{10}{20} = 0.85$	yes: B over C
3 (picked C)	$A = (10, 30)$	$\frac{10}{40} + \frac{30}{80} = 0.625$	yes: C over A
	$B = (10, 18)$	$\frac{10}{40} + \frac{18}{80} \approx 0.48$	yes: C over B

So these three choices contain **three contradictions, one for every pair**:

1. B **and** C**, directly.** B over C (line 2) and C over B (line 3): each was affordable when the other was picked. This is case 1 of section 7.3, on two lines.
2. A **and** C**, through the chain.** C over A comes straight from line 3. A over C is only *indirect*: A over B (line 1) and B over C (line 2). C was never affordable when A was picked (its cost share on line 1 is $\frac{35}{20} + \frac{10}{60} \approx 1.92$). The cycle $A \to B \to C \to A$ only shows when you follow the chain.
3. A **and** B**, through the chain.** A over B is direct (line 1, cost share 0.8). B over A is indirect: B over C (line 2) and C over A (line 3).

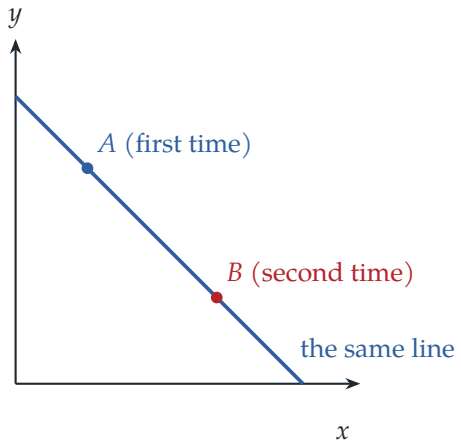
In a cycle, every pair contradicts: here one directly (B and C), the other two through the chain. Any one of them means the choices violate GARP. With two goods, as here, a cycle always contains a pair of lines that contradict each other directly (here B and C). Cycles in which no pair contradicts directly need three or more goods.

7.5 The same line twice

What if you face the *same* budget line twice and pick two different bundles, A the first time and B the second? Both cost exactly the budget, so each was affordable when the other was picked: A is revealed at least as good as B, and B at least as good as A. Together that says $A \sim B$: you are **indifferent** between them. That is allowed. GARP only forbids a cycle with a *strict* step (one bundle that was strictly cheaper than the budget), and here there is none.

So two different answers on the same line are **not** a contradiction under GARP: they show a tie, for example when your indifference curve runs along the budget line, as with perfect substitutes. They *are* a contradiction if we also assume that the best bundle is unique (strictly convex preferences, where only one point on the line touches the highest curve). Some studies use that stricter test (the strong axiom, SARP).

It becomes a clear violation as soon as one of the two picks is strictly inside the other's budget, which is exactly the two-line case 1 of section 7.3.



both cost the whole budget: a tie, not a cycle

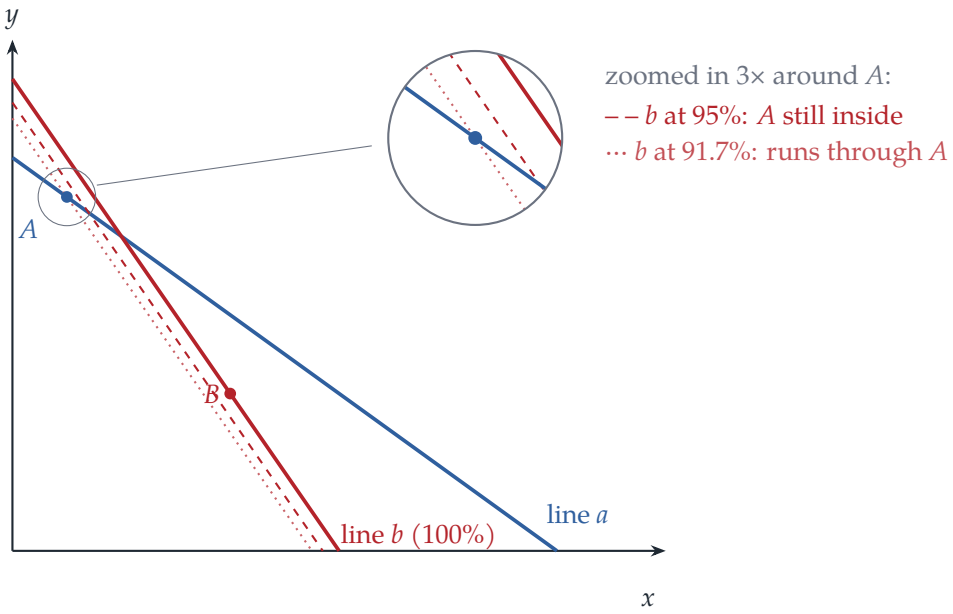
7.6 How close to consistent: the efficiency index in a picture

Go back to the contradiction of section 7.3, case 1: $A = (2, 18)$ on line a and $B = (8, 8)$ on line b. On line b you picked B, although A cost only 0.917 of that budget ($\frac{2}{12} + \frac{18}{24}$). So you paid the whole budget for B when A was available for 91.7% of it.

Now **shrink every budget line** toward the origin, to a fraction e of its money, and count a bundle as "revealed worse" only if it fits inside the shrunk line.

- At $e = 0.95$: line b shrinks to ends $(11.4, 22.8)$. A (cost share 0.917) is still inside: still "B over A", still a contradiction.
- At $e = 0.917$: the shrunk line b runs exactly through A. Just below that, A is outside: "B over A" disappears, and the contradiction with it.

The largest e at which no contradiction is left is the **critical cost efficiency index**: here CCEI $\approx$ 0.92. In words: your choices are consistent once we forgive you wasting up to about 8% of a budget. A fully consistent chooser has CCEI $= 1$; the further below 1, the more money the inconsistency "costs".



7.7 Afriat's theorem and how close to consistent

Afriat's theorem (1967), in words

A set of choices on budget lines passes GARP **if and only if** there is a well-behaved preference ordering (complete, transitive, more is better, and convex) under which every choice was the best affordable bundle.

This is a striking result. We never need to guess a formula for preferences: checking the choices for cycles is enough. It also means that convexity is an assumption we can never catch failing by watching choices on budget lines. Someone who loves extremes picks an end of the budget line, but so can someone with ordinary bowed-in curves who simply likes that good a lot: all we see is the picked point, so the two look the same. In general, whatever bundles a person picks, as long as the picks contain no cycle, we can always draw convex indifference curves (bowed in, possibly with some straight pieces) under which every pick is a best bundle on its line. So convexity is something we add because it makes the analysis simple, not something budget data could ever reject.

Real people are rarely perfectly consistent, so researchers also measure *how close* choices come, with the efficiency index (CCEI) drawn in section 7.6, the efficiency index in a picture: 1 means fully consistent, 0.95 means the violations are worth at most 5% of a budget. A useful benchmark: people who click on random points of many crossing lines almost always violate GARP, while most participants in budget experiments come close to full consistency.

7.8 The task in class

In class you make choices on a series of budget lines, one point on each line, with the same rules as in the experiments: each line meets the axes somewhere between 10 and 100, at least one end is 50 or more, and the lines cross often. Afterwards we check the class's choices for cycles. Try

to choose what you would really want on each line; there are no right answers.

8 Next lecture

Indifference curves rank bundles without numbers; section 6 gave them a first set of numbers (utility, ordinal). Next lecture we write utility functions as formulas. We will see why only the order of those numbers matters (*ordinal* utility, as opposed to *cardinal*), meet the standard utility functions, compute the MRS from marginal utilities, and bring back the budget line to find the best affordable bundle.

Summary

- The budget line's slope, p_x/p_y, is the relative price: the market's rate of exchange. It is not the MRS.
- Preferences rank bundles: $\succsim$ (at least as good), $\succ$ (strictly better), $\sim$ (indifferent).
- Four rules: completeness, transitivity (no cycles; otherwise a money pump), more is better (for goods), convexity (averages at least as good as extremes).
- An indifference curve joins equally good bundles. Curves slope down and further out is better (more is better), never cross (transitivity and more is better), and bow in (convexity).
- The MRS is how much y you would give up for one more x: the steepness of the curve at a bundle. It diminishes along a bowed-in curve. Near a point, the curve is close to its tangent line; the error grows with the size of the step.
- Perfect substitutes: straight lines, constant MRS, usually a corner. Perfect complements: L-shapes, fixed proportions. Both are weakly convex.
- Choices on budget lines reveal preferences. A choice that was made while another bundle was affordable reveals a preference; cycles violate GARP. Afriat: no cycles $\Leftrightarrow$ the choices can be explained by well-behaved preferences.

Further reading. Frank, *Microeconomics and Behavior*, chapter 3, pp. 63–69 (week 4 page, additional readings; the password is given in class).